Archimedes: An Economical Photonic Demonstration of the Hadamard Gate Using (LOQC) Linear Optical Quantum Computing and the KLM Protocol

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"The shortest path between two truths, in the real domain, passes through the complex domain -- Le chemin le plus court entre deux vérités, dans le domaine réel, passe par le domaine complexe." - Jacques Hadamard (Paris, 1965)

Abstract

Linear Optical Quantum Computing (LOQC) provides a promising approach to scalable, room-temperature quantum computation by encoding quantum information in photonic states and manipulating them using passive optical components. This research paper presents *Archimedes*, a low-cost, benchtop photonic quantum processor that demonstrates a single-qubit Hadamard gate using principles derived from the Knill–Laflamme–Milburn (KLM) protocol. The system employs a pulsed 650 nm laser diode, beam splitters, phase shifters, and an Arduino-based timing and detection architecture to generate and verify photonic superposition states. Although simplified for educational and proof-of-concept purposes, the experiment illustrates key LOQC concepts including dual-rail encoding, measurement-induced nonlinearity, and post-selection. The project validates the feasibility of low-cost photonic experiments and highlights challenges such as photon loss, mode mismatch, and the need for high-efficiency single-photon sources in future iterations.

Keywords: Linear Optical Quantum Computing; Photonic Quantum Computing; KLM Protocol; Hadamard Gate; Open-Source Quantum Hardware

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1. Introduction

The Hadamard gate, named after the French mathematician Jacques Hadamard, is a fundamental single-qubit operation in quantum computing that transforms computational

basis states into equal superpositions. In matrix form, the Hadamard operation is defined as:

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Applied to the computational basis states, it yields:

$$H \mid 0\rangle = \frac{\mid 0\rangle + \mid 1\rangle}{\sqrt{2}}, H \mid 1\rangle = \frac{\mid 0\rangle - \mid 1\rangle}{\sqrt{2}}$$

In photonic quantum systems, the Hadamard gate can be implemented naturally using a 50/50 beam splitter when information is encoded in either polarization or spatial (dual-rail) modes. Linear Optical Quantum Computing (LOQC) leverages such passive optical elements—beam splitters, phase shifters, and mirrors—combined with measurement and post-selection to perform quantum logic.

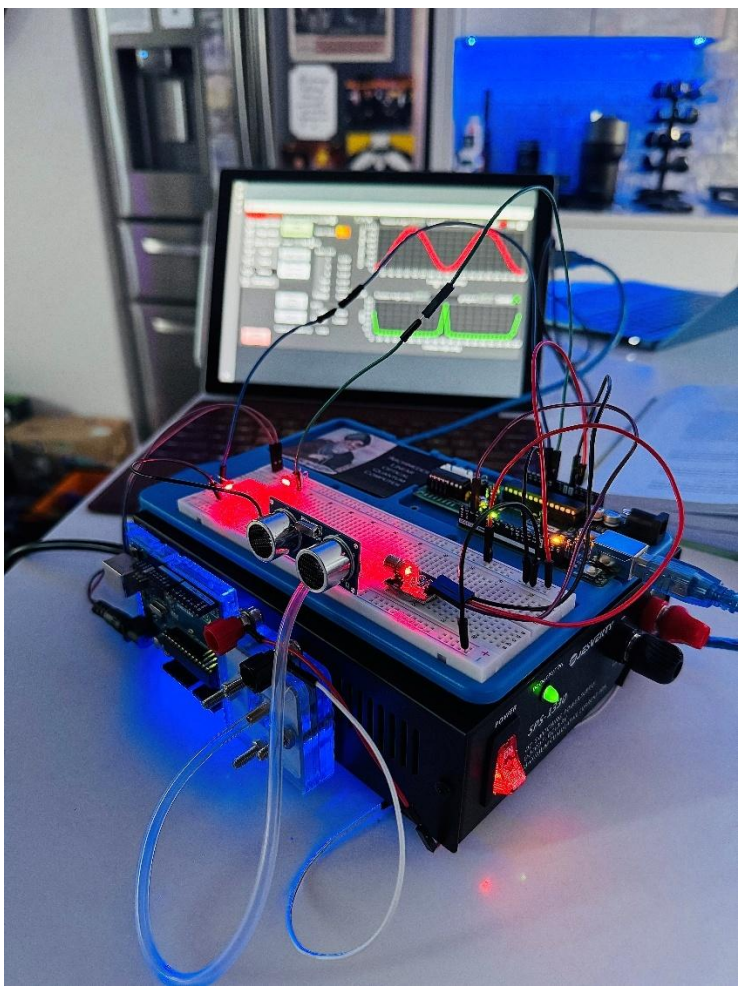

Figure 1. 'Archimedes' Photonic Linear Optical Quantum Computer

The Knill–Laflamme–Milburn (KLM) protocol demonstrated that scalable quantum computation is theoretically possible using only linear optics, ancillary photons, projective measurements, and classical feed-forward control. In the KLM framework, single-qubit gates like the Hadamard are straightforward with linear optics, while two-qubit gates (e.g., controlled-phase or CZ gates) are probabilistic and rely on measurement-induced nonlinearity. For a CZ gate in KLM, the operation can be represented as a parity check using beam splitters and ancillary modes, with success probability scaling with the number of ancillae. A simplified KLM CZ gate involves interfering target and control photons with ancilla photons on beam splitters, followed by post-selection on detection outcomes. The effective nonlinearity arises from conditioning on specific measurement results, such as:

$$|\psi\rangle = \alpha\,|00\rangle + \beta\,|01\rangle + \gamma\,|10\rangle + \delta\,|11\rangle \rightarrow \textit{post-selected on even parity detections}$$

While full KLM implementations require high-efficiency single-photon sources and fast optical switching, simplified demonstrations remain valuable for education, prototyping, and conceptual validation. This work introduces *Archimedes*, an economical and open-source photonic experiment designed to demonstrate a single-qubit Hadamard operation within the LOQC framework using readily available components. The system is intended as an accessible platform for students, researchers, and engineers exploring photonic quantum computing principles.

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2. System Design

2.1 Photon Source and Encoding

The Archimedes system uses a pulsed 650 nm laser diode as a pseudo-single-photon source. Pulse timing is controlled via an Arduino microcontroller to produce short, repeatable optical pulses suitable for interference measurements.

Quantum information is encoded using dual-rail path encoding, where logical states are represented by spatial modes:

$$|0\rangle = |1\rangle_{\text{upper}}\,|0\rangle_{\text{lower}}, |1\rangle = |0\rangle_{\text{upper}}\,|1\rangle_{\text{lower}}$$

This encoding allows straightforward implementation of single-qubit gates using beam splitters and mirrors.

2.2 Optical Layout

A 50/50 beam splitter serves as the core optical element implementing the Hadamard transformation. When a photon enters one input port, the beam splitter produces an equal

superposition of output paths, corresponding to the Hadamard operation on the dual-rail qubit:

$$|0\rangle \rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Photon detection is performed using photodiode sensor modules positioned at the output paths. While true coincidence counting is not implemented, detection events are time-correlated and post-selected to emulate simplified KLM-style measurement behavior.

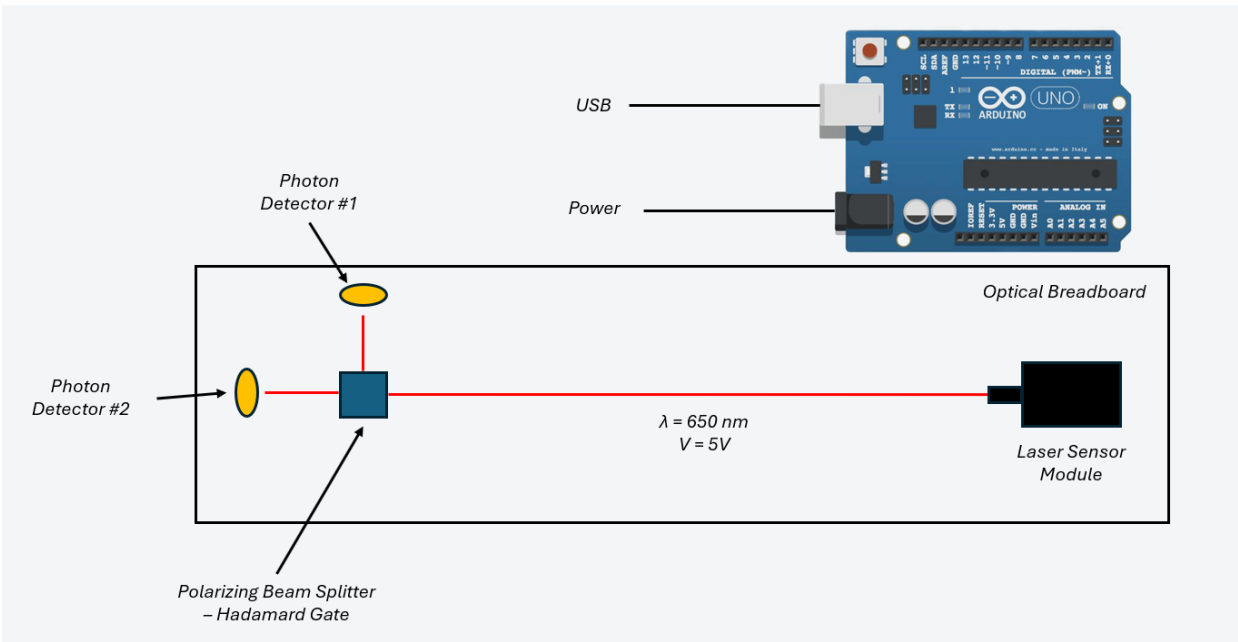

Figure 2. Optical layout of the Archimedes photonic Hadamard gate demonstration.

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3. Experimental Implementation

The experimental procedure consists of four primary stages:

1. Input State Preparation: A coherent optical pulse is injected into a single rail, representing a computational basis state (e.g., $|0\rangle$).
2. Hadamard Gate Application: The beam splitter transforms the input state into a superposition across the two spatial modes:

$$|0\rangle \rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

3. Post-Selection: Detection events consistent with expected interference behavior are selected to approximate KLM-style conditional operation.

4. Verification and Measurement: Path population statistics are recorded and analyzed to verify superposition behavior.

An Arduino microcontroller coordinates laser pulsing, sensor acquisition, and serial output. The embedded firmware generates randomized detection outcomes based on measured signal levels and outputs logical qubit states for inspection via a serial monitor.

```cpp
/*
 * 'Archimedes' Linear Optical Photonic Processor
 * Author: James Eckhart, Jr.
 * "Rise above oneself and grasp the world." - Archimedes
 */

const int TRIGGER_PIN = 2;   // Pulses the quantum circuit
const int H_PIN = A0;        // Measures horizontally polarized photons
const int V_PIN = A1;        // Measures vertically polarized photons
const int LED_PIN = 13;      // Onboard LED for visual feedback

void setup() {
  pinMode(TRIGGER_PIN, OUTPUT);
  pinMode(LED_PIN, OUTPUT);
  Serial.begin(9600);

  Serial.println("=== Archimedes Photonic Processor Initialized === FIRE THE LASER!!! -Dr. Evil");
}

int generateRandomBit() {
  // Pulse the laser
  digitalWrite(TRIGGER_PIN, HIGH);
  delay(100);
  digitalWrite(TRIGGER_PIN, LOW);

  // Read photon polarization levels
  float hValue = analogRead(H_PIN);
  float vValue = analogRead(V_PIN);

  // Determine random bit
  if (hValue > vValue) {
    digitalWrite(LED_PIN, LOW);  // Indicate bit 0
    return 0;
  } else if (hValue < vValue) {
    digitalWrite(LED_PIN, HIGH); // Indicate bit 1
    return 1;
  } else {
    // Equal photon counts — retry until a bit is generated
    return generateRandomBit();
  }
}

void loop() {
  int bit = generateRandomBit();
  Serial.print(" Quantum Bit -- Ket: |");
  Serial.print(bit);
  Serial.println(">");
  delay(1000); // Slight delay for readability
}
```

Figure 3. Arduino Microcontroller Code (C++)

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4. Results and Discussion

The experiment demonstrates clear interference behavior consistent with Hadamard gate operation in a photonic system, with expected balanced detection events between the two

logical states indicating successful generation of superposition states. The project highlights several practical challenges inherent to low-cost LOQC implementations, including optical loss, mode mismatch, detector inefficiency, and the absence of true single-photon sources. Nevertheless, the system effectively illustrates the core principles of photonic quantum logic and measurement-based operation.

While Archimedes does not implement a full KLM two-qubit gate, it provides a tangible foundation upon which more advanced photonic circuits—such as controlled-phase gates and entanglement sources—may be constructed.

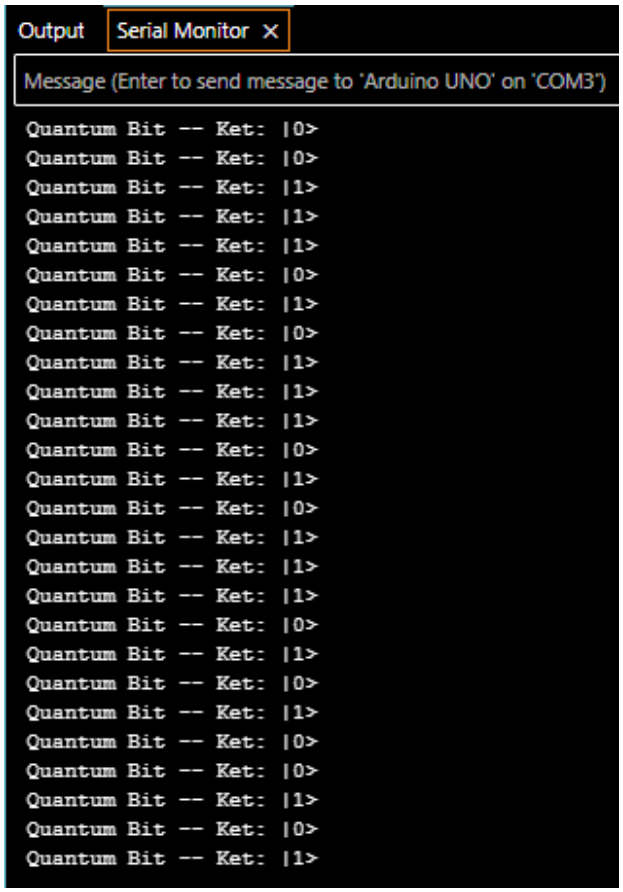

Figure 2. Quantum Bit Output from Serial Monitor -- the states |0⟩ (ket zero) and |1⟩ (ket one) represent the logical basis states of a qubit encoded in the dual-rail (path) encoding scheme.

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5. Conclusion

Archimedes provides a tangible, replicable demonstration of photonic quantum logic using the KLM framework. The project validates the feasibility of benchtop photonic quantum experiments and serves as an educational and prototyping platform for exploring linear optical quantum computing fundamentals.

All design files, firmware, schematics, Jupyter analysis notebooks, and supporting materials are released as open-source resources available at <https://github.com/JamesEckhartJr> to encourage replication, extension, and further experimentation in photonic quantum computing.

References

1. Knill, E., Laflamme, R., & Milburn, G. J. (2001). A scheme for efficient quantum computation with linear optics. Nature, 409, 46–52.
2. Additional references and implementation details are available in the project repository: <https://github.com/JamesEckhartJr>.